BUSINESS INTELLIGENCE LAB

EXPERIMENT 9

BI Security – Authentication, Authorization and Access Control using Microsoft Power BI

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1. Aim

To study and demonstrate Business Intelligence security by identifying users and roles, analysing authentication methods, designing authorization and access control, implementing Row-Level Security (RLS) in Microsoft Power BI Desktop, and assessing security risks with suitable controls.

4. Procedure

The procedure is organized into five parts covering 17 steps. Figures are numbered in order of appearance and are the same in the accompanying Practical Report. Screenshots are actual captures from the Power BI Desktop work done for this experiment.

Preliminary: Loading the Dataset

What was done: The Global Superstore Dataset.xlsx file was loaded into Power BI Desktop so that the Orders table (which contains the Region column used later for RLS) is available in the data model.

Power BI Desktop operations: Home → Excel workbook (or Get data → Excel) → select Global Superstore Dataset.xlsx → select the required sheets → Load.

Observation: The Data pane in the supplied screenshots lists the Orders, People and Returns tables. The Orders table includes fields such as Product ID, Product Name, Profit, Quantity, Region, Row ID, Sales, Segment, Ship Date, Ship Mode, Shipping Cost, State and Sub-Category.

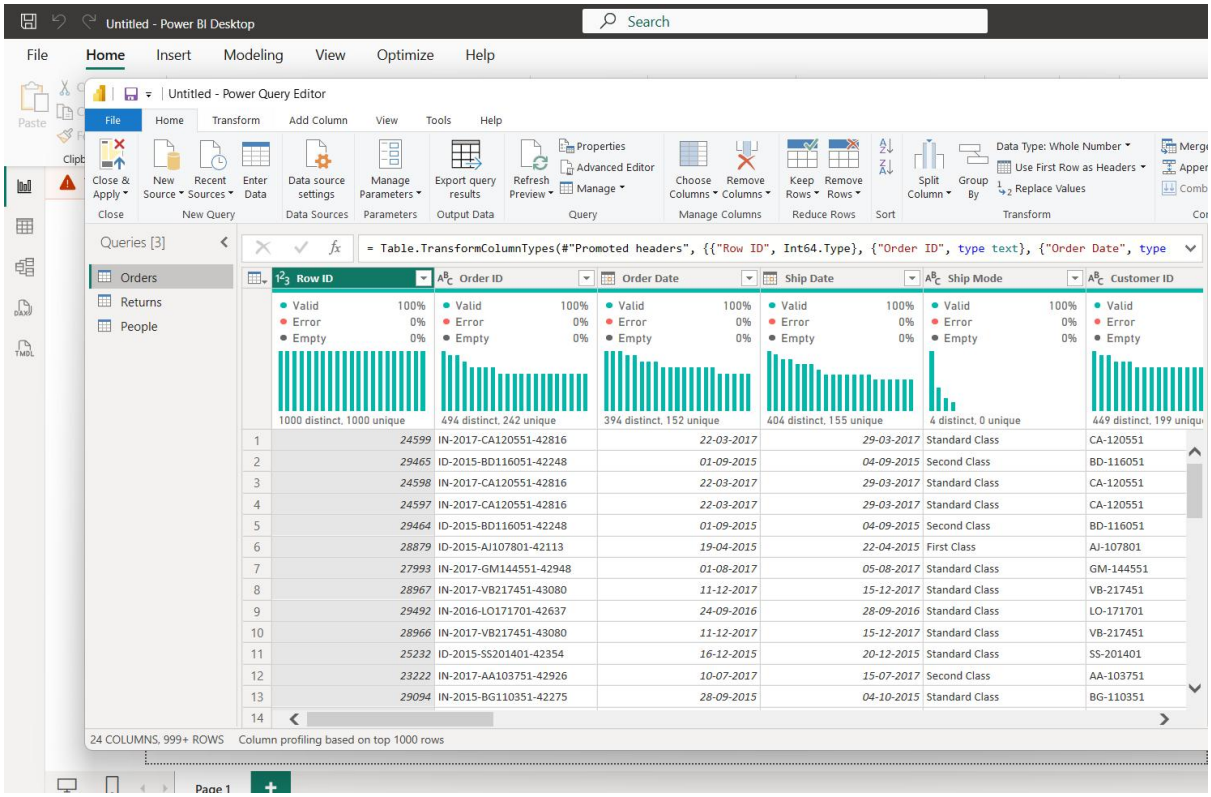


Figure 1: Global Superstore Dataset Loaded in Power BI Desktop

PART A: USER AND ROLE IDENTIFICATION

Step 1: Identify Organizational Users

What was done: The users who consume BI reports in the organization were identified: **CEO, Sales Manager, Finance Manager and Regional Manager**. These roles have different responsibilities, so each needs different reports and a different level of access. A fifth role, **BI Administrator**, is added in Part C because that role manages the BI environment.

Power BI Desktop operations: Home → Enter data → type the values in the first column and rename it User Roles. The table was then opened in Power Query Editor (Home → Transform data), where it appears as the query Role_Matrix.

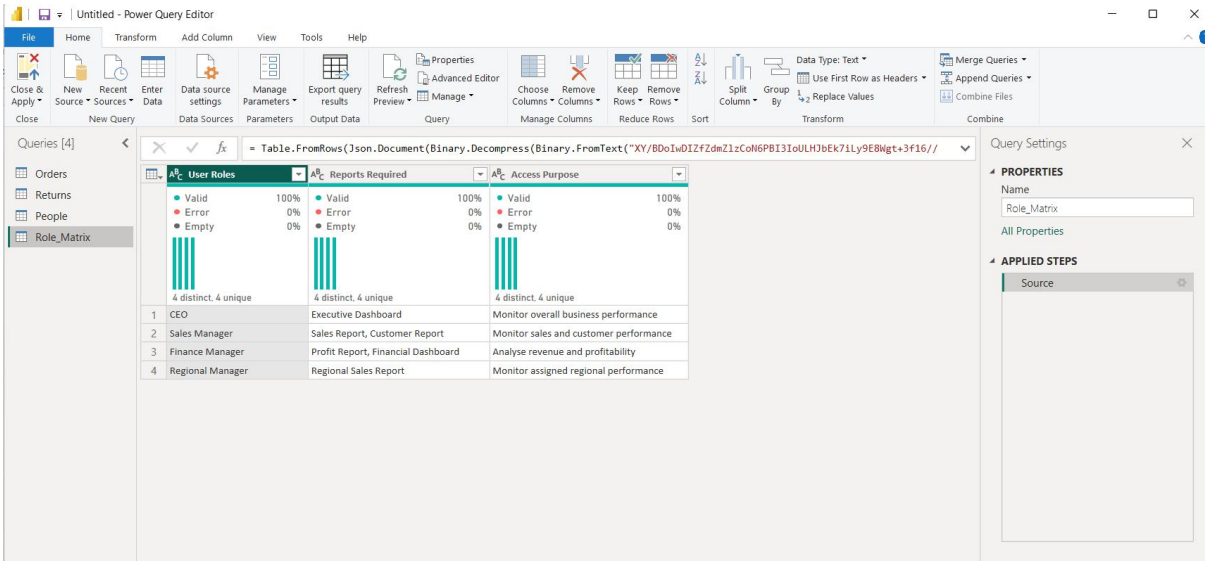


Figure 2: Role_Matrix Table in Power Query Editor

Observation: Power Query Editor shows the Role_Matrix query with the User Roles column holding four distinct, unique values (CEO, Sales Manager, Finance Manager, Regional Manager). The column profile shows 100% valid, 0% error and 0% empty values. The Source step formula indicates that the table was created using Enter data.

Step 2: Define Information Requirements

What was done: For each user role, the reports required and the purpose of access were defined, so that access can later be limited to what each role actually needs.

Power BI Desktop operations: The columns Reports Required and Access Purpose were added to the same Enter data table (Role_Matrix), as shown in Figure 2.

User Role	Reports Required	Access Purpose
CEO	Executive Dashboard	Monitor overall business performance
Sales Manager	Sales Report, Customer Report	Monitor sales and customer performance
Finance Manager	Profit Report, Financial Dashboard	Analyse revenue and profitability
Regional Manager	Regional Sales Report	Monitor assigned regional performance

Observation: Each role needs a distinct set of reports. For example, the Regional Manager needs only the Regional Sales Report, whereas the CEO requires the Executive Dashboard for overall business performance. This difference is the basis for the permissions and RLS design in the later parts.

Step 3: Create the Role Matrix

What was done: The Role_Matrix table (User Roles, Reports Required, Access Purpose) was loaded into the model and displayed in the report as a Role Matrix.

Power BI Desktop operations: Close & Apply → Report view → Visualizations pane → Table visual → drag Access Purpose, Reports Required and User Roles from the Role_Matrix table in the Data pane into the visual.

Access Purpose	Reports Required	User Roles
Monitor overall business performance	Executive Dashboard	CEO
Analyse revenue and profitability	Profit Report, Financial Dashboard	Finance Manager
Monitor assigned regional performance	Regional Sales Report	Regional Manager
Monitor sales and customer performance	Sales Report, Customer Report	Sales Manager

Figure 3: Role Matrix

Observation: The matrix consolidates the four roles with their reports and purpose in a single view, and is used as the reference for the Access Control Matrix in Part C.

PART B: AUTHENTICATION ANALYSIS

Step 4: Study Authentication Methods

What was done: Three authentication methods relevant to accessing Power BI were studied:

- **Username and Password:** the user signs in using login credentials.
- **Organizational Account:** the user accesses Power BI using the organization's Microsoft account.
- **Multi-Factor Authentication (MFA):** the user verifies identity using an additional authentication method on top of the password.

Power BI Desktop operations: Not applicable. This step is a study; the outcome is recorded in the comparison table in Step 5.

Step 5: Compare Username and Password, Organizational Account and MFA

What was done: The three methods were compared on their description and advantages and recorded in a table (Authentication_Methods) created using Enter data.

Power BI Desktop operations: Home → Enter data → enter the columns Authentication Method, Description and Advantages → name the table → Load → add a Table visual in Report view.

Advantages	Authentication Method	Description
Centralized identity management	Organizational Account	Users access Power BI using their organization's Microsoft account
Provides an extra layer of security	Multi-Factor Authentication (MFA)	Users verify their identity using an additional authentication method
Simple and easy to use	Username and Password	Users sign in using their login credentials

Figure 4: Authentication Methods Comparison

Authentication Method	Description	Advantages
Username and Password	Users sign in using their login credentials	Simple and easy to use
Organizational Account	Users access Power BI using their organization's Microsoft account	Centralized identity management
Multi-Factor Authentication (MFA)	Users verify their identity using an additional authentication method	Provides an extra layer of security

Observation: Username and Password is the simplest method but offers the least protection on its own. An Organizational Account centralizes identity management, and MFA adds an extra verification layer. Combining an organizational account with MFA gives stronger protection than a password alone.

Step 6: Document Advantages of Secure Authentication

What was done: The advantages of secure authentication were documented, as displayed in the report using a text box.

Power BI Desktop operations: Home → Insert → Text box → type the advantages and apply bullet formatting.



Figure 5: Advantages of Secure Authentication

- Prevents unauthorized users from accessing organizational data.
- Protects confidential business information.
- Reduces the risk of account misuse.
- Provides an additional security layer through MFA.
- Helps maintain secure access to reports and dashboards.

Observation: Secure authentication is the first line of defence: it ensures only verified users reach reports, which is a prerequisite for the authorization controls in Part C and the RLS in Part D.

PART C: AUTHORIZATION AND ACCESS CONTROL DESIGN

Step 7: Create the Access Control Matrix

What was done: An Access Control Matrix was created to define, for every user role, which of the four permissions is granted.

Power BI Desktop operations: Home → Enter data → columns User Role, View, Edit, Share, Admin → name the table Access_Control_Matrix → Load → add a Table visual in Report view.

Admin	Edit	Share	User Role	View
No	No	No	CEO	Yes
No	No	No	Regional Manager	Yes
No	Yes	No	Finance Manager	Yes
No	Yes	No	Sales Manager	Yes
Yes	Yes	Yes	BI Administrator	Yes

Figure 6: Access Control Matrix

User Role	View	Edit	Share	Admin
CEO	Yes	No	No	No
Sales Manager	Yes	Yes	No	No
Finance Manager	Yes	Yes	No	No
Regional Manager	Yes	No	No	No
BI Administrator	Yes	Yes	Yes	Yes

Observation: The matrix covers five roles. The BI Administrator role appears here (it is not in the Role Matrix) because it manages the BI environment.

Step 8: Identify View, Edit, Share and Admin Permissions

What was done: The four permission types used in the matrix were defined:

- **View:** open and read reports and dashboards.
- **Edit:** modify report content, such as visuals and measures.
- **Share:** give other users access to reports or dashboards.
- **Admin:** manage the BI environment, including user access.

Power BI Desktop operations: Not applicable; these are the column definitions of the Access_Control_Matrix table.

Step 9: Assign Permissions to Users

What was done: Permissions were assigned to each role in the matrix as per the table in Step 7 (Figure 6). All roles have View. The Sales Manager and Finance Manager additionally have Edit. The CEO and Regional Manager are view-only. Only the BI Administrator has Edit, Share and Admin together.

Observation: The assignment follows role-based access control and the principle of least privilege. It is recorded in the matrix only; it is not an actual permission setting in Power BI Service.

Step 10: Justify Permission Assignments:

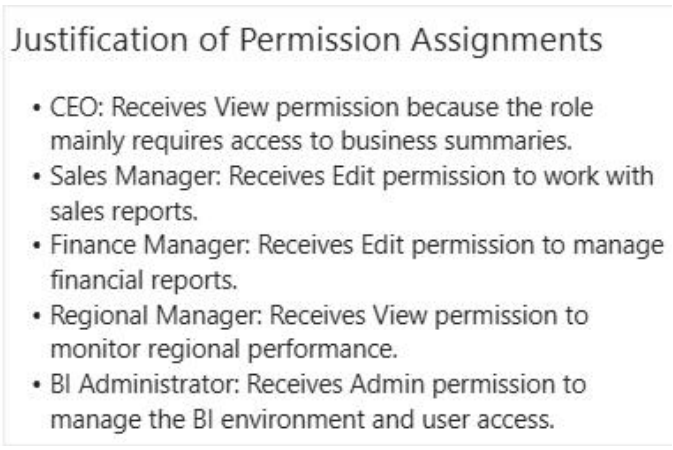


Figure 7: Justification of Permission Assignments

- **CEO:** Receives View permission because the role mainly requires access to business summaries.
- **Sales Manager:** Receives Edit permission to work with sales reports.
- **Finance Manager:** Receives Edit permission to manage financial reports.
- **Regional Manager:** Receives View permission to monitor regional performance.
- **BI Administrator:** Receives Admin permission to manage the BI environment and user access.

Observation: Each permission is tied to what the role actually does, so no role receives more access than its function requires.

PART D: ROW-LEVEL SECURITY (RLS) PLANNING

Step 11: Design the RLS Strategy

What was done: The RLS strategy was planned: managers should only see data for their own region, so access is restricted by filtering the **Region** column of the **Orders** table. One RLS role is defined per region, and the CEO, who monitors overall performance, is designated for all regions.

Power BI Desktop operations: Report view → Table visual with Region and Sales from the Orders table, to identify the regions present in the data.

Step 12: Prepare the RLS Design Matrix

What was done: The RLS Design Matrix was prepared to map each user role to its assigned region and data access.

Power BI Desktop operations: Home → Enter data → columns Assigned Region, Data Access, User Role → name the table RLS_Design → Load → Table visual in Report view (Page 2).



Assigned Region	Data Access	User Role
All Regions	All regional data	CEO
Eastern Asia	Eastern Asia only	Eastern Asia Manager
Eastern Europe	Eastern Europe only	Eastern Europe Manager
North Africa	North Africa only	North Africa Manager
South America	South America only	South America Manager

Figure 9: RLS Design Matrix

Assigned Region	Data Access	User Role
All Regions	All regional data	CEO
Eastern Asia	Eastern Asia only	Eastern Asia Manager
Eastern Europe	Eastern Europe only	Eastern Europe Manager
North Africa	North Africa only	North Africa Manager
South America	South America only	South America Manager

> ***Note:*** *Figure 9 is cropped from the screenshot taken during View As testing, which is why the RLS_Design table is visible there in full. The RLS_Design table is a documentation table; the actual security rules are the DAX filters in Step 13.*

Observation: Four of the five rows (Eastern Asia, Eastern Europe, North Africa, South America) correspond to RLS roles implemented in Step 13. The "All Regions" row for the CEO is documented in the matrix only; no CEO role appears in the Manage security roles screenshots.

Step 13: Explain and Document the Actual RLS Configuration in Power BI Desktop

This step documents what was actually configured. It covers three activities: creating roles (Manage roles), defining DAX filters, and testing with View As.

13.1 Creating Security Roles (Manage roles)

Power BI Desktop operations: Modeling tab → Security group → Manage roles → in the Roles pane click **New** → name the role → in the Tables pane select the **Orders** table → add the filter in the Rules pane.

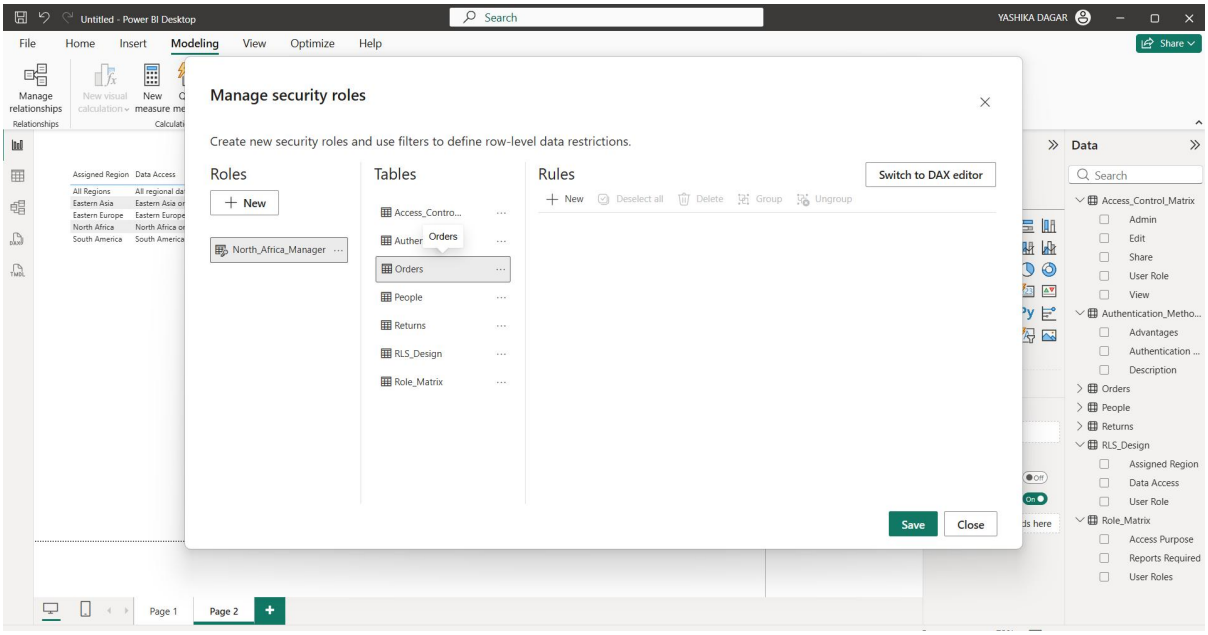


Figure 10: Manage Security Roles: Role Created and Orders Table Selected

Observation: The Manage security roles dialog shows the first role, North_Africa_Manager, created and the Orders table selected in the Tables pane. The tables available in the model are Access_Control_Matrix, Authentication_Methods, Orders, People, Returns, RLS_Design and Role_Matrix.

13.2 Defining DAX Filters

Power BI Desktop operations: In the Rules pane click **Switch to DAX editor** → enter the filter expression → click the tick mark to validate → **Save**. Power BI shows "Successfully applied role changes" after saving. The Orders table shows a filter icon once a rule is defined on it.

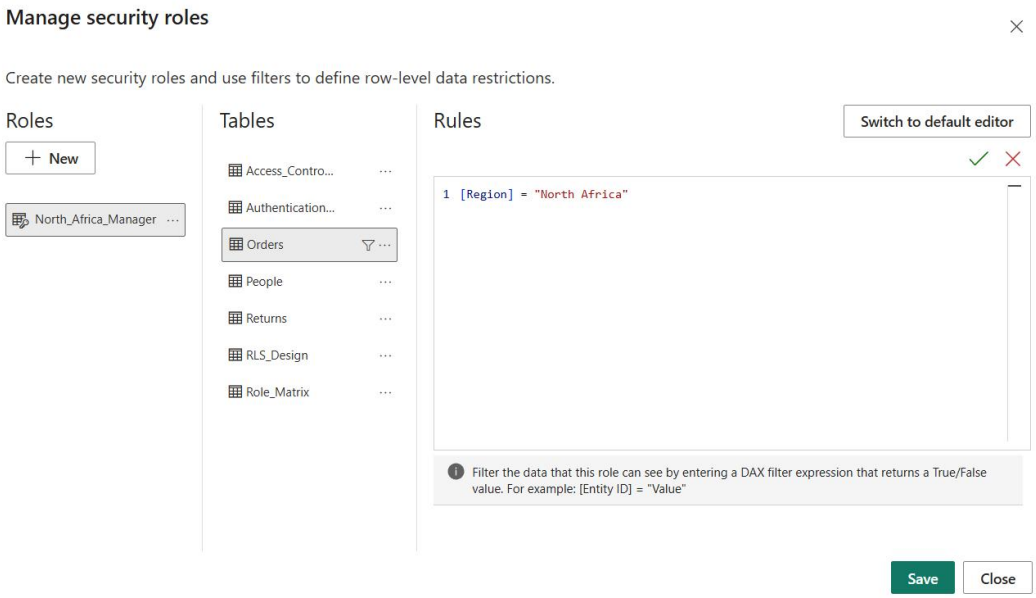


Figure 11: Row-Level Security Filter for North_Africa_Manager

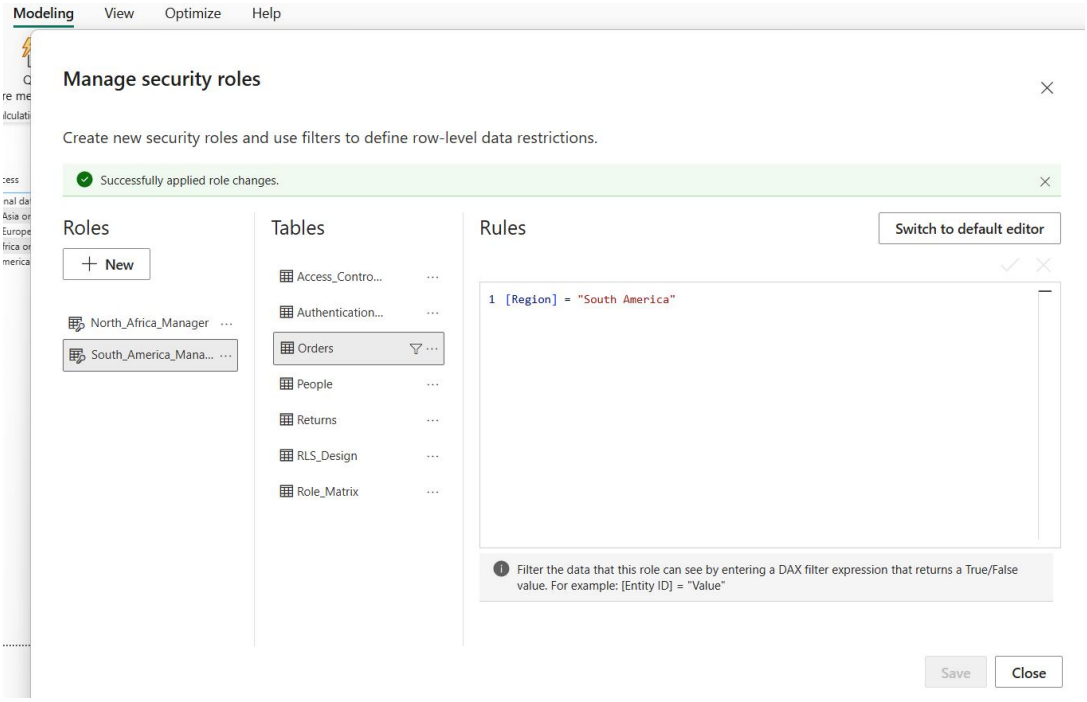


Figure 12: Row-Level Security Filter for South_America_Manager

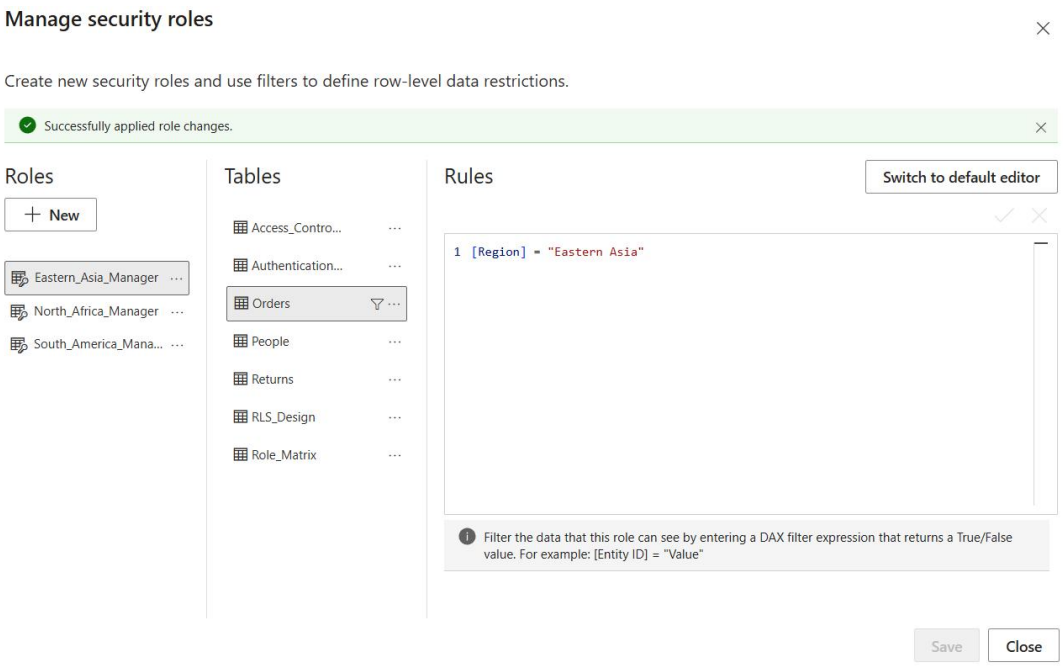


Figure 13: Row-Level Security Filter for Eastern_Asia_Manager

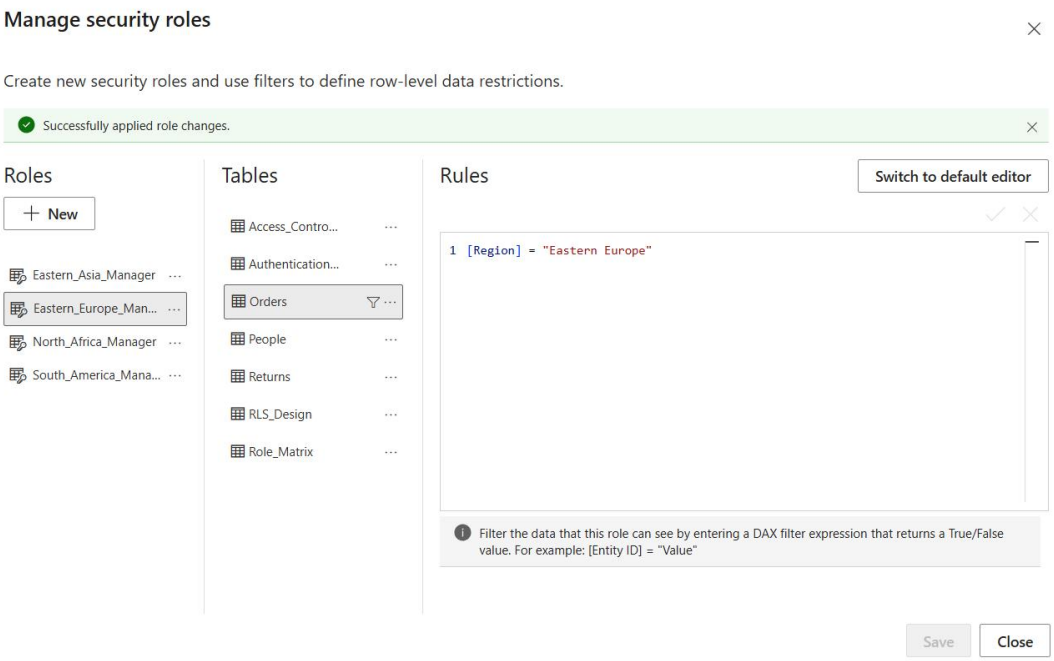


Figure 14: Row-Level Security Filter for Eastern_Europe_Manager

The four roles and their DAX filters, exactly as visible in the screenshots, are:

RLS Role	Table	DAX Filter Expression
Eastern_Asia_Manager	Orders	[Region] = "Eastern Asia"
Eastern_Europe_Manager	Orders	[Region] = "Eastern Europe"
North_Africa_Manager	Orders	[Region] = "North Africa"
South_America_Manager	Orders	[Region] = "South America"

Observation: Each role applies one table filter on the Region column of Orders, returning True only for rows of its own region. This is static RLS because the region value is typed directly into the expression.

13.3 Testing the Roles with View As

Power BI Desktop operations: Modeling tab → Security group → **View as** → tick one or more roles in the View as roles dialog (the dialog also offers None and Other user) → OK. A banner "Now viewing as" appears in Report view. Click **Stop viewing** to return to normal view.

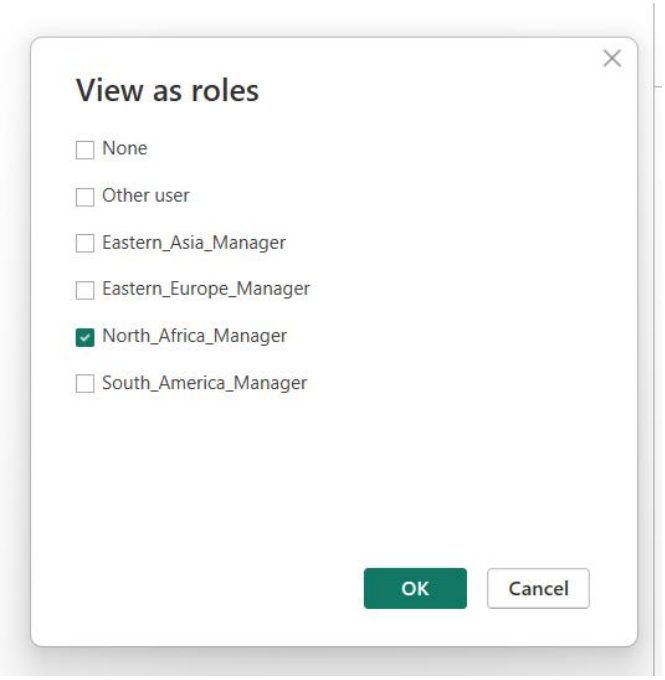


Figure 15: RLS Testing: View as Roles (North_Africa_Manager)

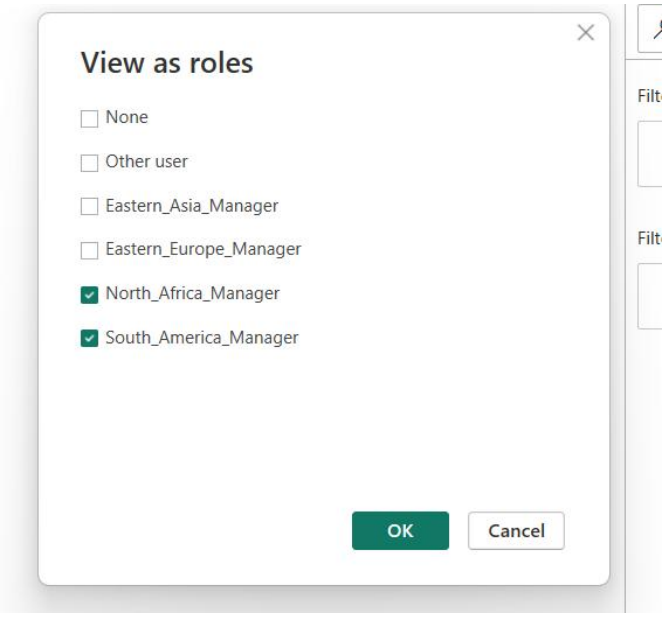


Figure 16: RLS Testing: View as Roles (North_Africa_Manager and South_America_Manager)

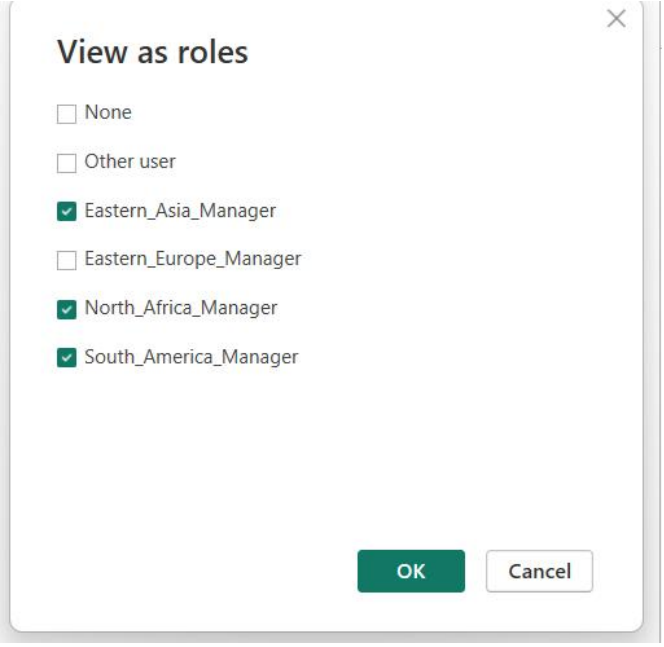


Figure 17: RLS Testing: View as Roles (Eastern_Asia_Manager, North_Africa_Manager and South_America_Manager)

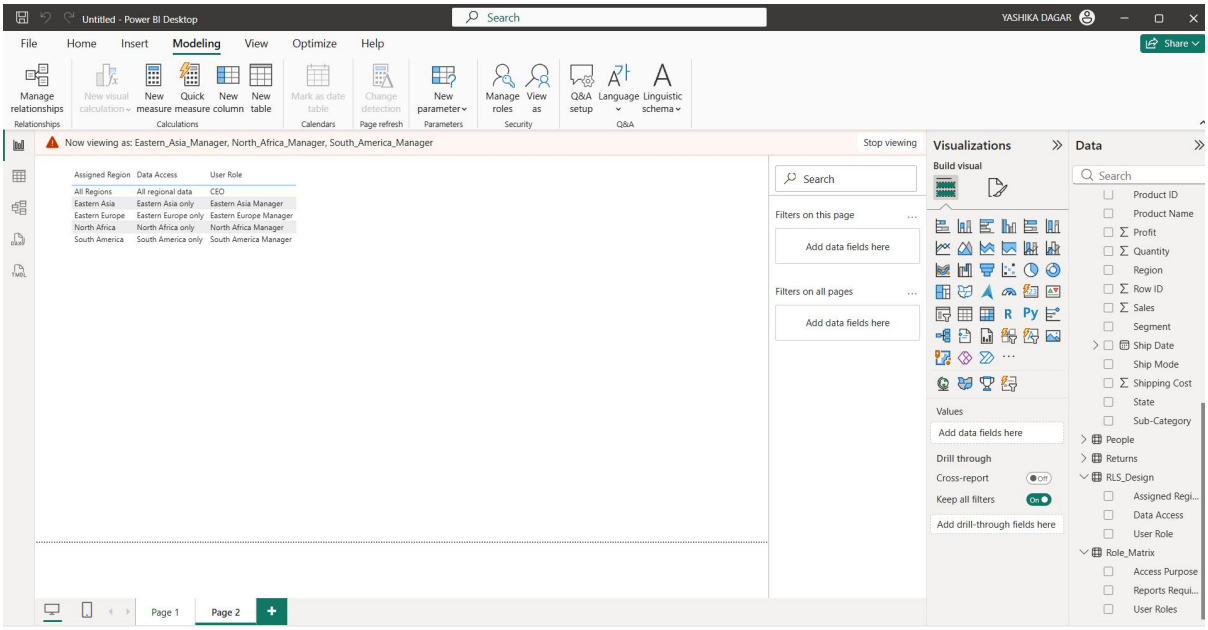


Figure 18: RLS Testing: "Now viewing as" Banner in Report View

Observation: The roles were selected in three combinations: North_Africa_Manager alone; North_Africa_Manager with South_America_Manager; and Eastern_Asia_Manager, North_Africa_Manager and South_America_Manager together. The final screenshot shows the banner "Now viewing as: Eastern_Asia_Manager, North_Africa_Manager, South_America_Manager". When several roles are selected together, Power BI combines their filters, so the user sees the rows allowed by any of the selected roles.

Step 14: Document the Security Benefits of RLS

What was done: The benefits of RLS were documented:

- Each manager sees only the data of the assigned region, which protects confidential information.
- One report can serve many users, so separate reports per region are not needed.
- It supports least-privilege access at the data (row) level, in addition to report-level permissions.
- Rules are defined centrally in the data model, which makes them easier to maintain and review.
- It reduces the risk of accidental exposure of other regions' data.

Observation: RLS complements authentication and the Access Control Matrix: authentication confirms identity, permissions control what a user may do, and RLS controls which rows the user may see.

PART E: SECURITY RISK ASSESSMENT

Step 15: Identify Security Risks

What was done: The main security risks in the BI environment were identified: **Unauthorized Access, Excessive Permissions, Data Leakage and Weak Authentication**. For each risk, the possible impact and a recommended control were recorded.

Power BI Desktop operations: Home → Enter data → enter the columns Security Risk, Possible Impact and Recommended Control → name the table Security_Risk_Matrix.

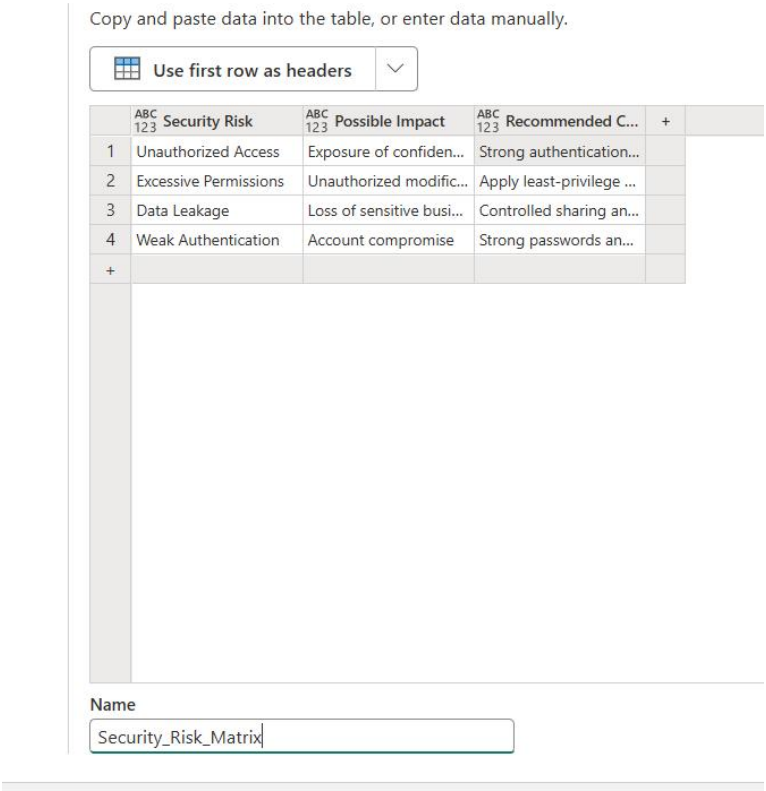


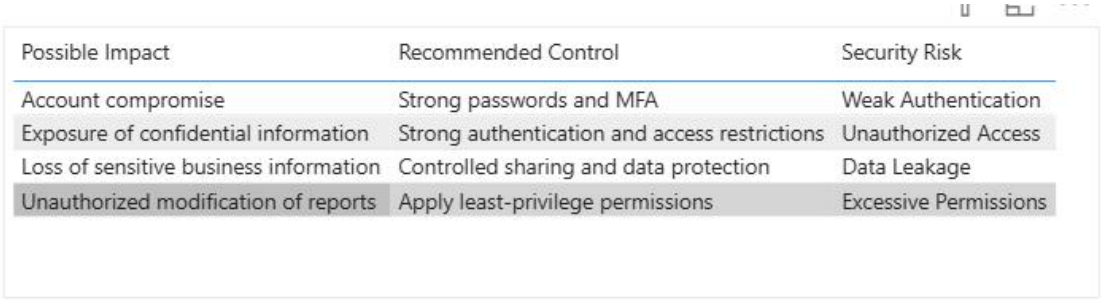
Figure 20: Creating the Security_Risk_Matrix Table Using Enter Data

Observation: The Enter data dialog shows four risk rows. The table name is entered as Security_Risk_Matrix.

Step 16: Prepare the Security Risk Assessment Matrix

What was done: The table was loaded and displayed in Report view as the Security Risk Assessment Matrix.

Power BI Desktop operations: Load → Report view → Table visual → add Security Risk, Possible Impact and Recommended Control.



Possible Impact	Recommended Control	Security Risk
Account compromise	Strong passwords and MFA	Weak Authentication
Exposure of confidential information	Strong authentication and access restrictions	Unauthorized Access
Loss of sensitive business information	Controlled sharing and data protection	Data Leakage
Unauthorized modification of reports	Apply least-privilege permissions	Excessive Permissions

Figure 21: Security Risk Assessment Matrix

Security Risk	Possible Impact	Recommended Control
Unauthorized Access	Exposure of confidential information	Strong authentication and access restrictions
Excessive Permissions	Unauthorized modification of reports	Apply least-privilege permissions
Data Leakage	Loss of sensitive business information	Controlled sharing and data protection
Weak Authentication	Account compromise	Strong passwords and MFA

Observation: Every risk is paired with its impact and a control, and the controls link back to earlier parts: authentication (Part B), least-privilege permissions (Part C) and restricted data access (Part D).

Step 17: Prepare Security Recommendations

What was done: Security recommendations were prepared from the Recommended Control column of the Security Risk Assessment Matrix (Figure 21):

- Use strong authentication and access restrictions to prevent unauthorized access.
- Apply least-privilege permissions so that users do not hold more access than their role needs.
- Use controlled sharing and data protection to prevent data leakage.
- Use strong passwords and MFA to reduce the risk of account compromise.
- Apply Row-Level Security so that each manager can see only the assigned region's data.

Observation: These recommendations are documented as proposed controls. MFA enablement and workspace permission settings were not configured as part of this experiment.

5. Result

The Business Intelligence security concepts were studied and applied using Microsoft Power BI Desktop with the Global Superstore dataset:

- Four organizational roles (CEO, Sales Manager, Finance Manager and Regional Manager) were identified, and a Role Matrix of reports and access purpose was prepared.
- Username and Password, Organizational Account and MFA were compared, and the advantages of secure authentication were documented. This was a documented analysis; MFA was not configured.
- An Access Control Matrix assigning View, Edit, Share and Admin permissions to five roles (including BI Administrator) was prepared, with justification. It is a design document; no workspace or sharing permissions were configured.
- An RLS Design Matrix was prepared, and four RLS roles (Eastern_Asia_Manager, Eastern_Europe_Manager, North_Africa_Manager and South_America_Manager) were created in Power BI Desktop, each with a DAX filter on [Region] in the Orders table.
- The roles were tested using View As in Power BI Desktop, and the "Now viewing as" banner was displayed. Assignment of users to roles in Power BI Service was not carried out as part of the supplied evidence.
- A Security Risk Assessment Matrix covering four risks, with impacts and controls, was prepared along with security recommendations.

Access Purpose	Reports Required	User Roles
Monitor overall business performance	Executive Dashboard	CEO
Analyse revenue and profitability	Profit Report, Financial Dashboard	Finance Manager
Monitor assigned regional performance	Regional Sales Report	Regional Manager
Monitor sales and customer performance	Sales Report, Customer Report	Sales Manager

Advantages	Authentication Method	Description
Centralized identity management	Organizational Account	Users access Power BI using their organization's Microsoft account
Provides an extra layer of security	Multi-Factor Authentication (MFA)	Users verify their identity using an additional authentication method
Simple and easy to use	Username and Password	Users sign in using their login credentials

Advantages of Secure Authentication

- Prevents unauthorized users from accessing organizational data.
- Protects confidential business information.
- Reduces the risk of account misuse.
- Provides an additional security layer through MFA.
- Helps maintain secure access to reports and dashboards.

Advantages of Secure Authentication

- Prevents unauthorized users from accessing organizational data.
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- Helps maintain secure access to reports and dashboards.

Justification of Permission Assignments

- CEO: Receives View permission because the role mainly requires access to business summaries.
- Sales Manager: Receives Edit permission to work with sales reports.
- Finance Manager: Receives Edit permission to manage financial reports.
- Regional Manager: Receives View permission to monitor regional performance.
- BI Administrator: Receives Admin permission to manage the BI environment and user access.

Admin	Edit	Share	User Role	View
No	No	No	CEO	Yes
No	No	No	Regional Manager	Yes
No	Yes	No	Finance Manager	Yes
No	Yes	No	Sales Manager	Yes
Yes	Yes	Yes	BI Administrator	Yes

Assigned Region	Data Access	User Role
All Regions	All regional data	CEO
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Exposure of confidential information	Strong authentication and access restrictions	Unauthorized Access
Loss of sensitive business information	Controlled sharing and data protection	Data Leakage
Unauthorized modification of reports	Apply least-privilege permissions	Excessive Permissions

Benefits of Row-Level Security

- Restricts access to confidential information.
- Prevents users from viewing unauthorized regional data.
- Provides role-based data visibility.
- Improves privacy and data protection.
- Reduces the chances of accidental data exposure.

Security Recommendations

- Use strong authentication methods.
- Enable Multi-Factor Authentication.
- Assign permissions according to user responsibilities.
- Implement Row-Level Security for regional data.
- Regularly review user access and workspace permissions.
- Restrict sharing of confidential reports.
- Monitor access and security-related activities.